Analytical Responses:

○ Why does fixed ε-greedy produce a linear cumulative regret curve O(T), whereas UCB produces a logarithmic regret curve O(log T)?

The ε-greedy algorithm explores with a constant value over the course of the entire experiment. The UCB algorithm on the other hand adapts its rate/probability of exploration over time as it becomes more confident in its probabilities. Thus, the cumulative regret for the ε-greedy algorithm increases with the number of trials/runs and the cumulative regret for the UCB algorithm becomes logarithmic.